Lecture 16: Power Series

Lecturer: Chng Zhi Yee (zy_chng@um.edu.my)

Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

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Learning Outcome

At the end of this lecture, you should be able to:

- Define a power series and identify its general form centered at a given point.
- Determine the interval and radius of convergence of a power series using appropriate convergence tests.
- Perform various operations on power series.

Power Series

So far we have studied about the infinite series of numbers, now we will look into the "infinite polynomials"

← polynomial with infinite degree.

A power series about $x = 0$ is a series of the form

$$\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + \cdots + c_n x^n + \cdots .$$

A power series about $x = a$ is a series of the form

↗ $a=0$

$$\sum_{n=0}^{\infty} c_n (x-a)^n = c_0 + c_1 (x-a) + c_2 (x-a)^2 + \cdots + c_n (x-a)^n + \cdots$$

where the center a and the coefficients $c_0, c_1, c_2, \ldots, c_n, \ldots$ are constants.

Power Series

Example: The geometric power series

$c_n = 1$ for all n.

$$\sum_{n=0}^{\infty} 1 x^n = 1 + x + x^2 + \cdots + x^n + \cdots$$

$a = 1$

geometric series with $r = x$

- convergent if $|r| = |x| < 1$
  - * convert to $\frac{1}{1-r} = \frac{1}{1-x}$

$\sum x^n = 1 + x + x^2 + \ldots = \frac{1}{1-x}$, valid for $|x| < 1$

Define a polynomial $P_n(x)$ be the partial sum of the series.

$P_n(x)$ will approach $\frac{1}{1-x}$ when $|x| < 1$

$P_0(x) = 1$

$P_1(x) = 1 + x$

$P_2(x) = 1 + x + x^2$

$\vdots$

$P_n(x) = 1 + x + x^2 + \ldots + x^n$

As n getting bigger, $P_n(x)$ will become closer to $\frac{1}{1-x}$ for $|x| < 1$.

Example: The power series

$$1 - \frac{1}{2}(x-2) + \frac{1}{4}(x-2)^2 + \cdots + \left(-\frac{1}{2}\right)^n (x-2)^n + \cdots$$

geometric series with $a=1$, $r = -\frac{1}{2}(x-2)$

The series converges when $|r| < 1$

$\frac{1}{2}|x-2| = |-\frac{1}{2}(x-2)| < 1$

$|x-2| < 2$

$0 < x < 4$

diverges when $|r| \geq 1$

$x \leq 0$, $x \geq 4$.

So for $0<x<4$, the series converges to $\frac{1}{1-(-\frac{1}{2}(x-2))} = \frac{2}{x}$

Polynomial approximation of $P_n(x) = \frac{2}{x}$ near $x=2$. ← the nearer the x to 2, the more accuracy of the approximation.

$P_0(x) = 1$

$P_1(x) = 1 - \frac{1}{2}(x-2) = 2 - \frac{x}{2}$

$P_2(x) = 1 - \frac{1}{2}(x-2) + \frac{1}{4}(x-2)^2$

$= 3 - \frac{3}{2}x + \frac{x^2}{4}$

⋮

Convergence of Power Series

We can test the convergence of a power series by using Ratio Test.

Example: $\sum_{n=1}^{\infty} \underbrace{(-1)^{n-1}\frac{x^n}{n}}_{u_n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$

$\lim_{n\to\infty}\left|\frac{u_{n+1}}{u_n}\right|$ $\begin{cases} <1, & \text{converges absolutely} \\ >1, & \text{diverges} \\ =1, & \text{inconclusive} \end{cases}$

Let $u_n = (-1)^{n+1}\frac{x^n}{n}$

$$\left|\frac{u_{n+1}}{u_n}\right| = \left|\frac{(-1)^{n}\frac{x^{n+1}}{n+1}}{(-1)^{n-1}\frac{x^n}{n}}\right| = \left|\frac{x^{n+1}}{n+1}\cdot\frac{n}{x^n}\right|$$

$$= \left|\frac{n}{n+1}\cdot x\right|$$

$$= \left(\frac{n}{n+1}\right)|x| \longrightarrow |x|$$

From the Ratio Test,
the series converges absolutely
when $|x|<1$ and diverges at $|x|>1$

How about when $|x|=1$? $(x=-1, 1)$

For $x = -1$, $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n} = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{(-1)^n}{n}$

$$= \sum_{n=1}^{\infty} (-1)^{2n+1} \frac{1}{n}$$

$$= -\sum_{n=1}^{\infty} \frac{1}{n} \quad \text{(divergent)}$$

divergent harmonic series.

For $x = 1$, $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n} = \sum_{n=1}^{\infty} (-1)^{n+1} \left(\frac{1}{n}\right)$ ← alternating harmonic series (convergent)

A.S. 7.

So, the power series converges at $-1 < x \leq 1$ and diverges elsewhere.

Convergence of Power Series

Example: $\sum_{n=1}^{\infty}(-1)^{n-1}\frac{x^{2n-1}}{2n-1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$

Let $u_n = (-1)^{n-1}\frac{x^{2n-1}}{2n-1}$

$$\left|\frac{u_{n+1}}{u_n}\right| = \left|\frac{(-1)^{n}\frac{x^{2(n+1)-1}}{2(n+1)-1}}{(-1)^{n-1}\frac{x^{2n-1}}{2n-1}}\right| = \left|\frac{x^{2n+1}}{2n+1}\cdot\frac{2n-1}{x^{2n-1}}\right| \quad (x^2)$$

$$= \left|\left(\frac{2n-1}{2n+1}\right)x^2\right|$$

$$= \left(\frac{2n-1}{2n+1}\right)x^2 \longrightarrow x^2$$

From the ratio test, the series converges when

$x^2 < 1 \Rightarrow -1 < x < 1$ and diverges when $x^2 > 1$

$(x < -1, \; x > 1)$

For $x = -1$, $\sum (-1)^{n-1} \frac{x^{2n-1}}{2n-1} = \sum (-1)^{n-1} \frac{(-1)^{2n-1}}{2n-1}$

$= \sum (-1)^n \left(\frac{1}{2n-1}\right)$

(convergent by Alternating Series Test)

For $x = 1$, $\sum (-1)^{n-1} \frac{x^{2n-1}}{2n-1} = \sum (-1)^{n-1} \frac{1}{2n-1}$

(convergent by AST)

So, the power series converges at $-1 \leq x \leq 1$ and diverges elsewhere.

Convergence of Power Series

Example: $\sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$

Let $u_n = \frac{x^n}{n!}$

$$\left|\frac{u_{n+1}}{u_n}\right| = \left|\frac{x^{n+1}}{(n+1)!} \cdot \frac{n!}{x^n}\right| = \frac{1}{n+1}|x| \to 0 < 1 \text{ for all } x.$$

So from Ratio Test, the series converges for all $x \in \mathbb{R}$.

Convergence of Power Series

Example: $\sum_{n=0}^{\infty} n!x^n = 1 + x + 2!x^2 + 3!x^3 + \cdots$

Let $u_n = n!\,x^n$

$$\left|\frac{u_{n+1}}{u_n}\right| = \left|\frac{(n+1)!\,x^{n+1}}{n!\,x^n}\right| = (n+1)|x| \to \infty \text{ for all } x \neq 0$$

($(n+1) \to \infty$)

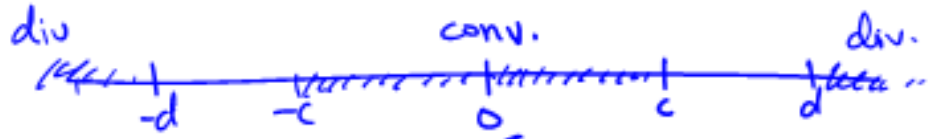
and when $x = 0$, $(n+1)|x| \to 0$.

From Ratio Test, the series converges at $x = 0$ and diverges everywhere else.

Convergence of Power Series

The Convergence Theorem for Power Series

If the power series



$$\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \cdots$$

converges at $x = c \neq 0$, then it converges absolutely for all x with $|x| < |c|$. (If the series diverges at $x = d$, then it diverges for all x with $|x| > |d|$.)

① ②

Proof

① Suppose that $\sum a_n x^n$ converges at $x = c \neq 0$

then $\lim_{n\to\infty} a_n c^n = 0$ (or else by the nth term test, it would be divergent)

Then there exists N such that for all $n \geq N$, $|a_n c^n| < 1$

Convergence of Power Series

$$\Rightarrow |a_n| < \frac{1}{|c|^n}$$

Now take any x such that $|x| < |c| \Rightarrow \frac{|x|}{|c|} < 1$

For all $n \geq N$, $|a_n x^n| = |a_n|\,|x^n| < \frac{1}{|c|^n}|x^n|$

$$= \left(\frac{|x|}{|c|}\right)^n$$

$$< 1$$

Now, since $\sum \left(\frac{|x|}{|c|}\right)^n$ is a geometric series with ratio $\frac{|x|}{|c|} < 1$ (convergent).

By DCT, $\sum |a_n x^n|$ also converges. $\Rightarrow \sum a_n x^n$ converges.

② Suppose $\sum a_n d^n$ diverges.

Let $x > d$.

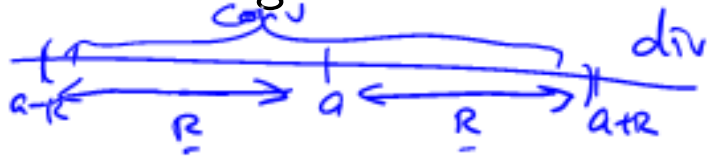
If $\sum a_n x^n$ converges, then by ①, $\sum a_n d^n$ would have to be convergent (contradiction)

So $\sum a_n x^n$ must diverge.

Radius of Convergence

The convergence of the series $\sum c_n(x-a)^n$ is described by one of the following three cases:

1. There is a positive number R such that the series diverges for x with $|x-a| > R$ but converges absolutely for x with $|x-a| < R$. The series may or may not converge at either of the endpoints $x = a-R$ and $x = a+R$.



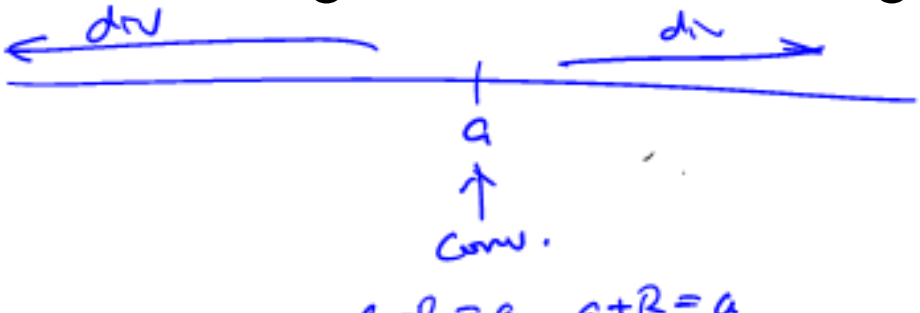
At the endpoints $a-R$ and $a+R$, we have to test seperately.

2. The series converges absolutely for every x $(R = \infty)$.



Converges everywhere.

3. The series converges at $x = a$ and diverges elsewhere $(R = 0)$.



diverges everywhere except at $x=a$.

$a-R=a$, $a+R=a$

R: Radius of convergence.

Radius of Convergence

$\sum_{n=0}^{\infty} c_n (x-a)^n$

Case 2

converges absolutely for all x.

$(R = \infty)$

Case 3

diverges everywhere except $x=a$

$(R=0)$

Case 1

converges on some interval & radius R.

- converges absolutely at $|x-a|<R$
- diverges at $|x-a|>R$
- may or may not converges at either of the endpoints : $x = a-R$, $x = a+R$

Radius of Convergence

Substitute $x = a - R$ $\left(\sum a_n (-R)^n\right)$

and $x = a + R$ $\left(\sum a_n (R)^n\right)$

and then check the convergency.

Interval of convergence : interval of radius centred at $x = a$ where the power series converges.

Case 1

- $(a-R, a+R)$
- $[a-R, a+R]$
- $(a-R, a+R]$
- $[a-R, a+R)$

Case 2 : $x \in \mathbb{R}$ $(-\infty, \infty)$

Case 3 : $x = a$.

Interval of Convergence

How to test a power series for convergence?

- Use Ratio Test (or Root Test) to find the largest open interval where the series converges absolutely,

$$|x - a| < R$$

- If R is finite, test for convergence or divergence at each endpoint. Use a Comparison Test, the Integral Test, or the Alternating Series Test.
- If R is finite, the series diverges for $|x - a| > R$.

Interval of Convergence

Example: Find the interval of convergence for $\sum_{n=1}^{\infty} \frac{x^n}{n}$. ← $a=0$.

$a=0$

Apply Ratio Test to find $|x-a|<R$

Let $u_n = \frac{x^n}{n}$

$$\left|\frac{u_{n+1}}{u_n}\right| = \left|\frac{x^{n+1}}{n+1} \cdot \frac{n}{x^n}\right| = \frac{n}{n+1}|x| \to |x|$$

The series converges at $|x|<1$ and diverges at $|x|>1$.

At the endpoints: when $x=-1$, $\sum \frac{(-1)^n}{n}$ is a convergent alternating harmonic series.

when $x=1$, $\sum \frac{1}{n}$ is a divergent harmonic series.

So the series converges at $[-1, 1)$.

Interval of Convergence

Example: Find the interval of convergence for $\sum_{n=1}^{\infty} \frac{2^n(x-2)^n}{n(3^n+1)}$.

$a=2$

Apply Ratio Test to find $|x-a|<R$.

Let $u_n = \dfrac{2^n (x-2)^n}{n(3^n+1)}$

$$\left|\frac{u_{n+1}}{u_n}\right| = \frac{2^{n+1}(x-2)^{n+1}}{(n+1)(3^{n+1}+1)} \cdot \frac{n(3^n+1)}{2^n(x-2)^n}$$

$$= \left|\frac{2(x-2)n(3^n+1)}{(n+1)(3^{n+1}+1)}\right|$$

$$= \left(\frac{2n}{n+1}\right)\left(\frac{3^n+1}{3^{n+1}+1}\right)|x-2| \longrightarrow (2)\left(\tfrac{1}{3}\right)|x-2|$$

$\frac{2n}{n+1} \to 2$, $\frac{3^n+1}{3^{n+1}+1} \to \frac{1}{3}$

So, the series converges at $\frac{2}{3}|x-2|<1$

$$\Rightarrow |x-2|<\frac{3}{2} \leftarrow R.$$

and diverges at $|x-2|>\frac{3}{2}$

At the endpoints:

when $x-2=\frac{-3}{2}$, $\sum \frac{2^n(x-2)^n}{n(3^n+1)}$

$$=\sum \frac{2^n\left(\frac{-3}{2}\right)^n}{n(3^n+1)} \qquad \frac{2^n}{n(3^n+1)}\left(\frac{-3}{2}\right)^n$$

$$=\sum \frac{2^n(-1)^n 3^n}{n(3^n+1)2^n}$$

$$=\sum (-1)^n \frac{3^n}{n(3^n+1)}$$

Let $a_n = \frac{3^n}{n(3^n+1)}$

① $a_n > 0$ for all $n \geq 1$

② $a_n \geq a_{n+1}$ for all $n \geq 1$

③ $\lim_{n\to\infty} \frac{3^n}{n(3^n+1)} = \lim_{n\to\infty} \frac{3^n}{n(3^n)} = \lim_{n\to\infty} \frac{1}{n} = 0.$

By the Alternating Series Test, the series converges at $x-2 = \frac{-3}{2}$.

When $x-2 = \frac{3}{2}$, $\sum \frac{2^n (x-2)^n}{n(3^n+1)}$

$$= \sum \frac{2^n \left(\frac{3}{2}\right)^n}{n(3^n+1)}$$

$$= \sum \frac{3^n}{n(3^n+1)} \quad \begin{matrix} \to 3^n \\ \to n(3^n) \end{matrix}$$

Let $b_n = \frac{3^n}{n(3^n+1)}$ and $c_n = \frac{3^n}{n(3^n)} = \frac{1}{n}$

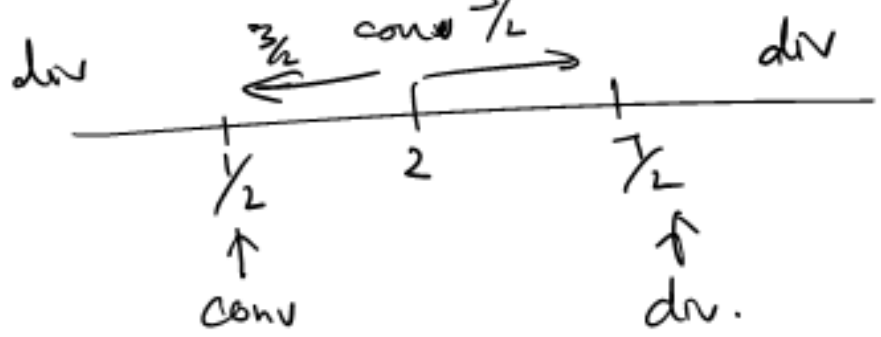
$$\lim_{n\to\infty} \frac{b_n}{c_n} = \lim_{n\to\infty} \frac{\frac{3^n}{n(3^n+1)}}{1/n} = \lim_{n\to\infty} \frac{3^n}{3^n+1} = 1$$

Since $\sum \frac{1}{n}$ is a divergent harmonic series,

by LCT, $\sum \frac{3^n}{n(3^n+1)}$ diverges.

So the series converges at $-\frac{3}{2} \le x-2 < \frac{3}{2}$,

so it converges at $\frac{1}{2} \le x < \frac{7}{2}$



Operations on Power Series

Series Multiplication for Power Series

If $A(x) = \sum_{n=0}^{\infty} a_n x^n$ and $B(x) = \sum_{n=0}^{\infty} b_n x^n$ converge absolutely for $|x| < R$, and

coefficient of the product.

$$c_n = a_0 b_n + a_1 b_{n-1} + a_2 b_{n-2} + \cdots + a_{n-1} b_1 + a_n b_0 = \sum_{k=0}^{n} a_k b_{n-k},$$

then $\sum_{n=0}^{\infty} c_n x^n$ converges absolutely to $A(x)B(x)$ for $|x| < R$

$$\left(\sum_{n=0}^{\infty} a_n x^n\right)\left(\sum_{n=0}^{\infty} b_n x^n\right) = \sum_{n=0}^{\infty} c_n x^n.$$

Eg: $A(x) = \sum_{n=0}^{\infty} x^n$, $B(x) = \sum_{n=0}^{\infty} x^n$ ($a_n, b_n = 1$ for all n)

↓

geometric series $a = 1$, $r = x$ converges if $|x| < 1$ to $\frac{1}{1-x}$

For $|x| < 1$, $A(x)$ and $B(x)$ both converges to $\frac{1}{1-x}$

Let $C(x) = A(x)B(x) = \sum c_n x^n$

where $c_n = a_0 b_n + a_1 b_{n-1} + \ldots + a_n b_0$

$= (1)(1) + (1)(1) + \ldots + (1)(1)$

$= n+1$

So $C(x) = \sum_{n=0}^{\infty} (n+1) x^n$ converges absolutely at $|x| < 1$ to $\left(\frac{1}{1-x}\right)\left(\frac{1}{1-x}\right) = \frac{1}{(1-x)^2}$

Operations on Power Series

Theorem: If $A(x) = \sum_{n=0}^{\infty} a_n x^n$ converges absolutely for $|x| < R$ and f is a continuous function, then $\sum_{n=0}^{\infty} a_n (f(x))^n$ converges absolutely on the set of points x where $|f(x)| < R$.

Example: $1/(1-4x^2) = \sum_{n=0}^{\infty} (4x^2)^n$

Recall that the power series $\sum_{n=0}^{\infty} x^n$ converges to $\frac{1}{1-x}$ at $|x| < 1$.

Let $f(x) = 4x^2$, f is continuous over $\mathbb{R}$.

So $\sum_{n=0}^{\infty} (f(x))^n = \sum_{n=0}^{\infty} (4x^2)^n$ converges to $\frac{1}{1-(4x^2)}$ at $|4x^2| < 1 \Rightarrow |x| < \frac{1}{2}$.

Term-by-Term Differentiation

If $\sum_{n=0}^{\infty} c_n(x-a)^n$ has radius of convergence $R > 0$, it defines a function

$$f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n \quad \text{on the interval} \quad a - R < x < a + R.$$

This function f has derivatives of all orders inside the interval, and we obtain the derivatives by differentiating the original series term by term:

$$f'(x) = \sum_{n=1}^{\infty} nc_n(x-a)^{n-1}, \frac{d}{dx}(x-a) = 1$$

$$f''(x) = \sum_{n=2}^{\infty} n(n-1)c_n(x-a)^{n-2},$$

and so on. Each of these derived series converges at every point of the interval $a - R < x < a + R$.

Operations on Power Series

Example: Find series for $f'(x)$ and $f''(x)$ if

$$f(x) = \frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots + x^n + \cdots = \sum_{n=0}^{\infty} x^n, \quad -1 < x < 1.$$

converges at $-1 < x < 1$.

$$f'(x) = \frac{1}{(1-x)^2} = (1-x)^{-2} = 1 + 2x + 3x^2 + \ldots + nx^{n-1} + \ldots = \sum_{n=0}^{\infty} nx^{n-1}$$

converges at $-1 < x < 1$.

$$f''(x) = -2(1-x)^{-3}(-1) = \frac{2}{(1-x)^3} = 2 + 6x + \ldots n(n-1)x^{n-2} + \ldots = \sum_{n=0}^{\infty} n(n-1)x^{n-2}$$

converges at $-1 < x < 1$

Ex: Find $f'''(x)$.

Term-by-Term Integration

Suppose that $(x) = \sum_{n=0}^{\infty} c_n(x-a)^n$ converges for $a - R < x < a + R$ $(R > 0)$. Then

$$\sum_{n=0}^{\infty} c_n \frac{(x-a)^{n+1}}{n+1}$$

converges for $a - R < x < a + R$ and

$$\int f(x)\,dx = \sum_{n=0}^{\infty} c_n \frac{(x-a)^{n+1}}{n+1} + C$$

for $a - R < x < a + R$.

Operations on Power Series

Example: Identify the function

$$f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots, \quad -1 < x < 1.$$

$f'(x) = 1 - x^2 + x^4 - x^6 + \cdots$, $-1 < x < 1$

which is a geometric series with $a = 1$, $r = -x^2$

For $|x| < 1$, $f'(x) = \frac{1}{1-(-x^2)} = \frac{1}{1+x^2}$

$|r| = |-x^2| < 1 \Rightarrow |x| < 1$

$f(x) = \int f'(x)\,dx = \int \frac{1}{1+x^2}\,dx = \arctan x + C$

Observe that $f(0) = 0$, so $f(0) = \arctan(0) + C = 0 \Rightarrow C = 0$

So $f(x) = \arctan x$ for $-1 < x < -1$.

Operations on Power Series

Consider the series

$$\frac{1}{1+t} = 1 - t + t^2 - t^3 + \cdots, \quad -1 < t < 1.$$

Notice that $\int_0^x \frac{1}{1+t}\,dt = \left[\ln|1+t|\right]_0^x$

$$= \ln(1+x) - \ln(1) = \ln(1+x)$$

So $\ln(1+x) = \int_0^x \frac{1}{1+t}\,dt = \int_0^x 1 - t + t^2 - t^3 + \ldots\,dt$

$$= \left[t - \frac{t^2}{2} + \frac{t^3}{3} - \frac{t^4}{4} + \ldots\right]_0^x$$

$$= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots$$

$$= \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1} \quad , -1 < x \leq 1$$